

ITAR-CONTROLLED TECHNICAL DATA

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CATEGORY 9 – AEROSPACE AND PROPULSION

ITEM 9A991 – Aircraft and related equipment

TECHNICAL DATA SUMMARY:

- Next-generation turbine blade cooling geometry (proprietary design)
- Hypersonic scramjet inlet design parameters
- Thrust vectoring nozzle control algorithms (v2.4)

DETAILED SPECIFICATIONS:

1. Turbine Blade Cooling Channels

- Material: Inconel 718
- Cooling Hole Diameter: 0.012 ± 0.001 inches
- Film Cooling Effectiveness: $\eta = 0.68$ @ Mach 0.85
- Internal Rib Configuration: Chevron Pattern Type III

2. Scramjet Inlet Design (Mach 6+)

- Compression Ramp Angle: 12.5°
- Shock Train Length: 18.4 inches
- Capture Area Ratio: 1.47

3. Thrust Vectoring Nozzle Parameters

- Max Deflection Angle: $\pm 18^\circ$
- Actuator Response Time: < 45 ms
- Control Law: PID with gain scheduling

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